

# SIMATS ENGINEERING

## ASSESSMENT TOOL 3

**COURSE NAME: INTERNET PROGRAMMING**  
**COURSE CODE: CSA4305**

---

### **COURSE OUTCOME COVERED**

**CO 4:** Formulate data-driven web solutions using XML, XSLT, and JSP within the MVC framework.  
(L4)

Assessment Tool: Class Test

Weightage: 10%

---

### **QUESTIONS:**

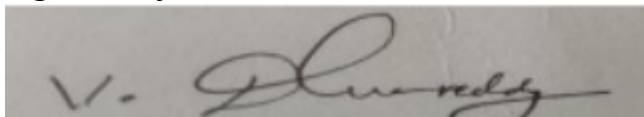
**Total Marks: 20**

1. Explain the fundamental concepts used in web development by defining the following terms with suitable explanations: **XML, JSP, MVC architecture, XSLT, XML Schema, JSTL, PHP, Database Connectivity, and Parser**. Your answer should clearly describe the purpose and role of each concept in web applications.
2. Explain how the above technologies are interconnected in building a web application. Describe how XML is structured and validated, how JSP and PHP are used for server-side processing, how MVC architecture organizes the application, and how database connectivity and parsers contribute to data processing and communication.

### ***Code of Conduct:***

*I \_\_V.Dharanireddy(192311219)\_\_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

### ***Signature of the student:***



**RUBRICS FOR CALCULATION:**

| <b>Criteria</b>                   | <b>Weightage</b> | <b>Level 4 – Exemplary</b>                                     | <b>Level 3 – Proficient</b>             | <b>Level 2 – Developing</b>          | <b>Level 1 – Beginning</b>                    |
|-----------------------------------|------------------|----------------------------------------------------------------|-----------------------------------------|--------------------------------------|-----------------------------------------------|
| <b>Concept Understanding</b>      | 25%              | Clearly explains all concepts with complete accuracy and depth | Minor gaps in understanding             | Basic understanding with some errors | Incorrect or very limited understanding       |
| <b>Content Coverage</b>           | 25%              | Covers all required topics comprehensively                     | Covers most topics with minor omissions | Covers some topics only              | Very few topics covered                       |
| <b>Technical Accuracy</b>         | 20%              | All explanations are technically correct and precise           | Minor technical errors                  | Some incorrect or vague explanations | Major technical errors                        |
| <b>Integration of Concepts</b>    | 15%              | Effectively connects and relates concepts logically            | Some connections made with minor gaps   | Limited connections between concepts | No clear relationship between concepts        |
| <b>Clarity &amp; Presentation</b> | 15%              | Well-structured, clear, and easy to understand                 | Mostly clear with minor issues          | Somewhat unclear or poorly organized | Poorly structured and difficult to understand |

19) Fundamental Concepts used in web development:

1. XML - Extensible Markup Language:

→ XML is a markup language used to store, organize and transport structured data.

Unlike HTML, which mainly describes how information should be displayed, what the data means.

Example:

```
<student>
  <id>101</id>
  <name>Ravi</name>
  <course>BCA</course>
  <mark>85</mark>
</student>
```

Here, the tags such as <student>, <name>, and <mark> describe the information.

Features of XML:

- Allows developers to create their own tags.
- Is platform and language independent
- Is human-readable.

Role in web applications:

- Exchange data between client and server
- Store application configuration.

2. JSP - Java Server Pages:

⇒ JSP is Java-based technology used to create dynamic web pages on server side.

⇒ A JSP page can contain HTML along with server-side

### Example:

```
<html>
<body>
  <h1> welcome to the student portal </h1>
  <p> Student Name: $ {student.name} </p>
</body>
</html>
```

∴ The server processes the JSP and sends resulting HTML to browser.

### How it works:

(Browser)

↓ (HTTP request)

(Web server / JSP container) (Process JSP) → (Generated HTML) → (Browser)

### Role in web applications:

- Creating dynamic web pages
- Displaying database information
- processing and presenting application data.

### 3. MVC Architecture:

MVC stands for model-view Controller. consists of 3 components;

A. Model

B. View

C. Controller

#### MVC Flow

(User) → (HTTP request) → Controller

(HTTP response) ← View  
↓ (User.)  
↑ (database)



XSLT - Extensible Stylesheet language Transformations  
⇒ A language used to transform XML documents in other formats.

→ commonly used to transform XML into;

→ HTML

→ Another XML documents and plaintext.

Example:

XML:

```
<student>
  <name>Ravi</name>
  <mark>85</mark>
</student>
```

HTML:

```
<h1> student details </h1>
  <p>name: Ravi</p>
  <p>mark: 85</p>
```

Basic process:

XML document → XSLT Stylesheet → HTML/XML/Text

Data → XML.

Transformation / presentation rules → XSLT

JSTL:

⇒ collection of standard tags used in JSP pages to perform common tasks without writing large amounts of Java.

Ex:

```
<c:forEach var="Student" items="{students}">
```

```
<p>{student.name}</p>
```

```
</c:forEach>
```

Examples include:

```
<c:if>
```

```
<c:choose>
```

```
<c:forEach>
```

```
<c:out>
```

Role in web applications:

→ JSTL functions like;

1. Condition statements
2. Loops & iterations
3. Formatting ....

### PHP - PHP: Hypertext Preprocessor:

⇒ It is a popular server-side scripting language used to develop dynamic websites and web applications.

#### Example:

```
<?php  
$name = "Ravi";  
echo "Welcome" . $name;  
?>
```

#### Database Connectivity:

⇒ It is connectivity refers to methods and technologies used by web application to communicate database.

⇒ It is commonly used to store - user accounts, products, orders, students records etc.

C-create, R-read, U-update, D-delete.

#### Parser:

⇒ A parser is software component that reads input according to particular syntax or structure and converts into application.

#### XML parser:

<student>

<name>Ravi</name>

<mark>85</mark>

</student>

XML → Extensible markup language store data.

JSP → Java Server pages, server-side web pages.

MVC → Model-view Controller, separate data, logics.

XSLT → XML Language Transform XML to HTML, XML

XML Schema → XML structure, define rules and datatypes.

JSTL → JSP standard - simplifies JSP.

PHP → Server side scripting language.

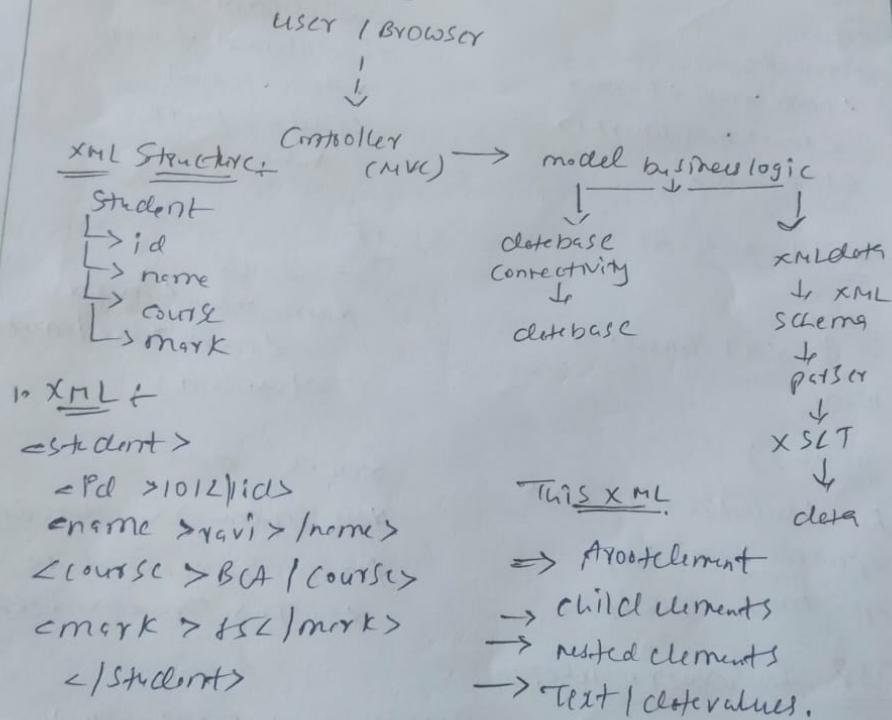
language used to applications.

Inter connection of XML, JSP, PHP, MVC, XSLT, XML Schema, JSTL, Parser...?

Solution:

- A modern web application consists of several layers network together to receive user request process.
- The technologies mentioned above are;

Simplified Flowchart:



1. XML :-

```

<student>
  <id>1012</id>
  <name>Ravi</name>
  <course>BCA</course>
  <mark>75</mark>
</student>
  
```

Ex:-

```

<Student id = "101">
  <name>Ravi</name>
  <course>BCA</course>
</Student>
  
```

(XML)

```

=> <Student>
  <name>Ravi</name>
</Student>
=> (properly nested)
  
```



### XML Validation:

Student  
→ id → integer  
→ name → string  
→ course → string  
→ mark → integer

### 3. parsers and XML

#### XML:

Ex:

```
<students>
  <name>ravi</name>
  <mark>85</mark>
</students>
```

### XSLT and XML

XML

↓ XSLT Transform

HTML

(HTML:)

```
<h1> Student details </h1>
<p name = ravi </p>
<p> mark = 85 </p>
```

### JSP for Server-Side:

Browser



web server



GET/Students

### Validation process:

XML document



XML Schema



validation

→ valid → apply

→ invalid → error.

### The parser identify:

Root element → student

Child element → name

value → Ravi

Child → mark

value → 85

Ex:

```
<students>
```

```
  <name>ravi</name>
```

```
  <mark>85</mark>
```

```
</students>
```

XML → data

XML Schema → validate data

Parser → reads data

XSLT → Transform data.

Browser → JSP/Controller ↓

database ← business logic



browser ←

data → JSP → HTML

## JSP in JSP:

(JSP)

```
<c:forEach var="Student" items="${Students}">
  <p> ${Student.name} </p>
</c:forEach>
```

(model data)

↓  
(JSP) → (JST) → (HTML)

## PHP Server:

Browser  
↓ → HTTP request.  
Web server  
↓  
PHP application  
↓ → database  
generated HTML  
↓  
Browser.

## MVC Architecture:

(Browser → Controller → model → database → a  
result → model → JSP/JST → Browser.)

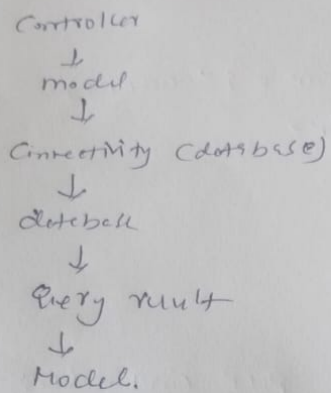
## PHP Controller:

- Form submissions
- Authentication
- Sessions
- Business logic
- API requests.

## Advantages:

- ⇒ Easy to understand
- testing
- modify
- maintain
- scalable.

### Database Connectivity:



⇒ Supports

{ INSERT  
DELETE  
UPDATE  
DELETE }

### HOW DBLT and MVC WORK

